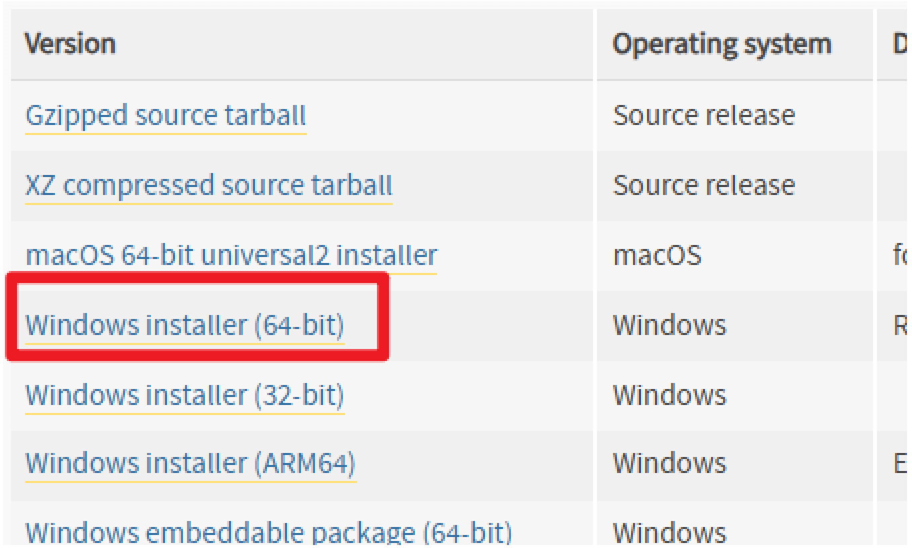
환경 설정

1. Python 설치

- Python 다운로드 페이지 에서 Python 3.12.10 설치 파일 받기



Version	Operating system	D
Gzipped source tarball	Source release	
XZ compressed source tarball	Source release	
macOS 64-bit universal2 installer	macOS	f
Windows installer (64-bit)	Windows	R
Windows installer (32-bit)	Windows	
Windows installer (ARM64)	Windows	E
Windows embeddable package (64-bit)	Windows	

- 설치 시 `add pyhton.exe to PATH` 체크

설치 후 확인

```
python --version
```

```
Windows PowerShell
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PS C:\Users\jongy> python --version
Python 3.12.10
PS C:\Users\jongy> where.exe python
C:\Users\jongy\AppData\Local\Programs\Python\Python312\python.exe
C:\Users\jongy\AppData\Local\Programs\Python\Python311\python.exe
C:\Users\jongy\AppData\Local\Microsoft\WindowsApps\python.exe
PS C:\Users\jongy>
```

2. 가상 환경

- `git bash` 로 진행

- 파일 탐색기로 실습 폴더(빈 폴더)생성한다.
- 해당 폴더를 열고 마우스 오른쪽 버튼으로 `Git Bash here` 클릭
- 가상 환경 설치

```
python -m venv .venv
```

- 가상 환경 활성화

```
source .venv/Scripts/activate
```

```
jongy@DESKTOP-JONG MINGW64 /c/WORKSPACE/ai_2_lecture
$ python -m venv .venv

jongy@DESKTOP-JONG MINGW64 /c/WORKSPACE/ai_2_lecture
$ source .venv/Scripts/activate
((.venv) )
jongy@DESKTOP-JONG MINGW64 /c/WORKSPACE/ai_2_lecture
$
```

3. 패키지 설치

- 먼저 `pip` 버전 업그레이드

```
python -m pip install --upgrade pip
```

```
$ python -m pip install --upgrade pip
Requirement already satisfied: pip in c:\workspace\ai_2_lecture\.venv\lib\si
ackages (25.0.1)
Collecting pip
  Downloading pip-26.2-py3-none-any.whl.metadata (4.6 kB)
Downloading pip-26.2-py3-none-any.whl (1.8 MB)
   ---------------------------------------- 1.8/1.8 MB 11.0 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 25.0.1
    Uninstalling pip-25.0.1:
      Successfully uninstalled pip-25.0.1
Successfully installed pip-26.2
((.venv) )
jongy@DESKTOP-JONG MINGW64 /c/WORKSPACE/ai_2_lecture
$ 
```

- `requirements.txt` 현재 폴더에 복사해서 설치

```
pip install -r requirements.txt
```

- 설치 후 확인

```
python -m pip check
```

> 예상 출력은 `No broken requirements found.`

```
((.venv) )
jongy@DESKTOP-JONG MINGW64 /c/WORKSPACE/ai_2_lecture
$ python -m pip check
No broken requirements found.
```

- 패키지 버전 확인

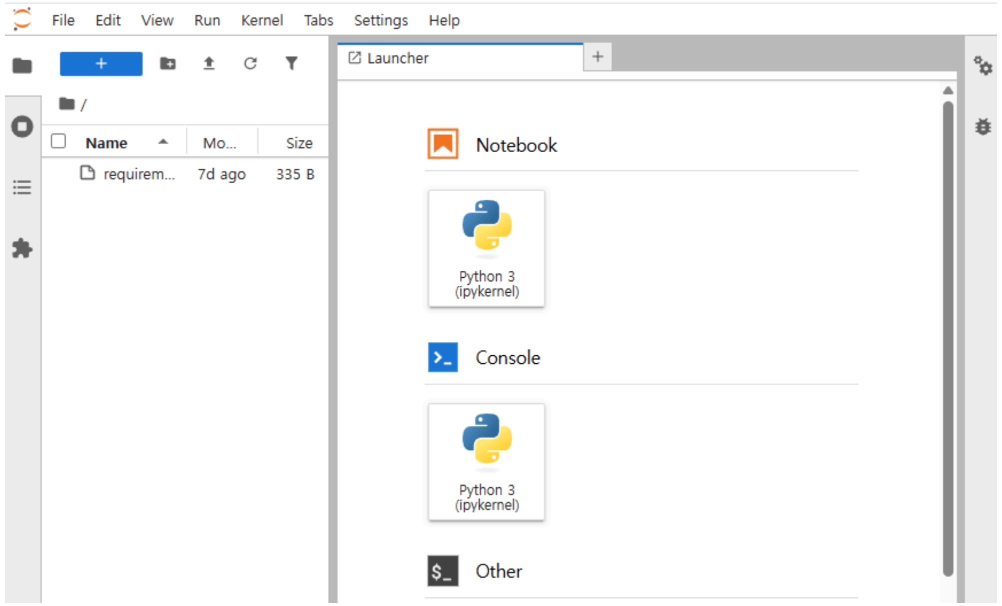
```
python -c "import requests, pandas, jupyterlab;
print('requests', requests.__version__);
print('pandas', pandas.__version__);
print('jupyterlab', jupyterlab.__version__)"
```

```
jongy@DESKTOP-JONG MINGW64 /c/WORKSPACE/ai_2_lecture
$ python -c "import requests, pandas, jupyterlab; print('requests', requests
.__version__); print('pandas', pandas.__version__); print('jupyterlab', jupy
terlab.__version__)"
requests 2.34.2
pandas 3.0.5
jupyterlab 4.6.2
((.venv) )
jongy@DESKTOP-JONG MINGW64 /c/WORKSPACE/ai_2_lecture
$
```

4. Jupyter Lab 실행

```
python -m jupyter lab

# 또는 현재 폴더가 아닌 특정 폴더 열기
python -m jupyter lab Day01/배포용
```



- 빈 폴더에서 실행한 경우 노트북 파일 또는 실습 폴더를 복사한다.